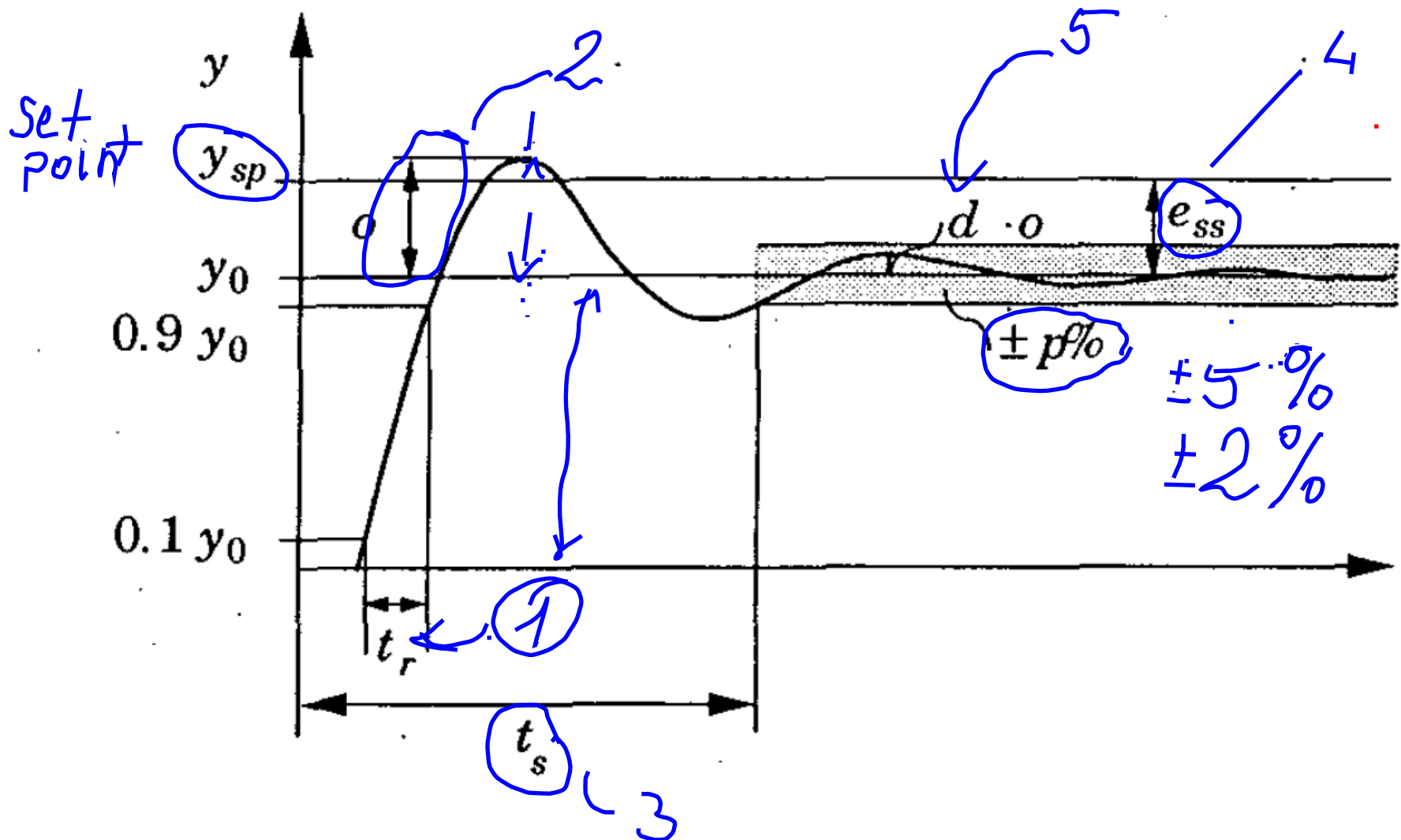


Sectional parameters of the step response



1. Rise Time: the time it takes for the plant output y to rise beyond 90% of the desired level for the first time.
2. Overshoot: how much the the peak level is higher than the steady state, normalized against the steady state.
3. Settling Time: the time it takes for the system to converge to its steady state.
4. Steady-state Error: the difference between the steady-state output and the desired output.

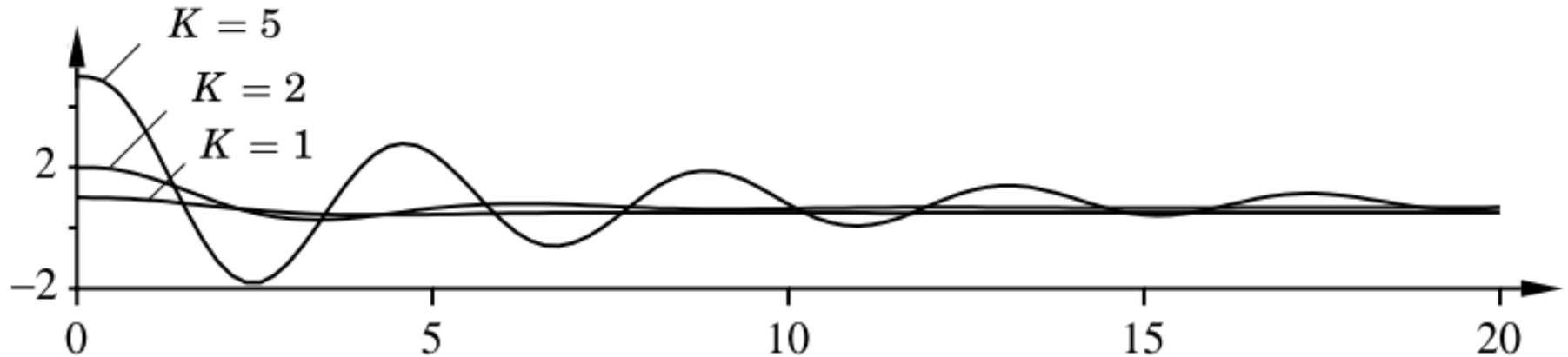
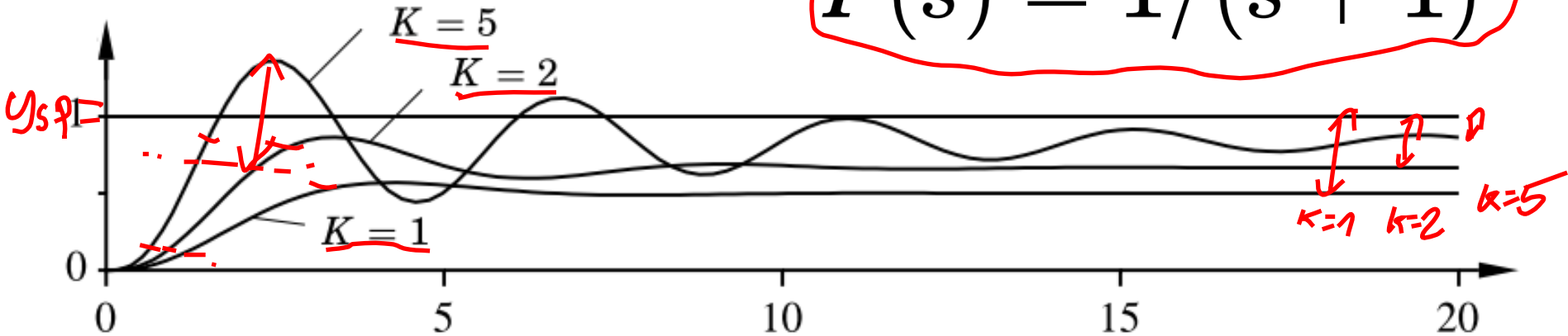
5. Damping

P Controller

$$G_c = K$$



$$P(s) = 1/(s + 1)^3$$

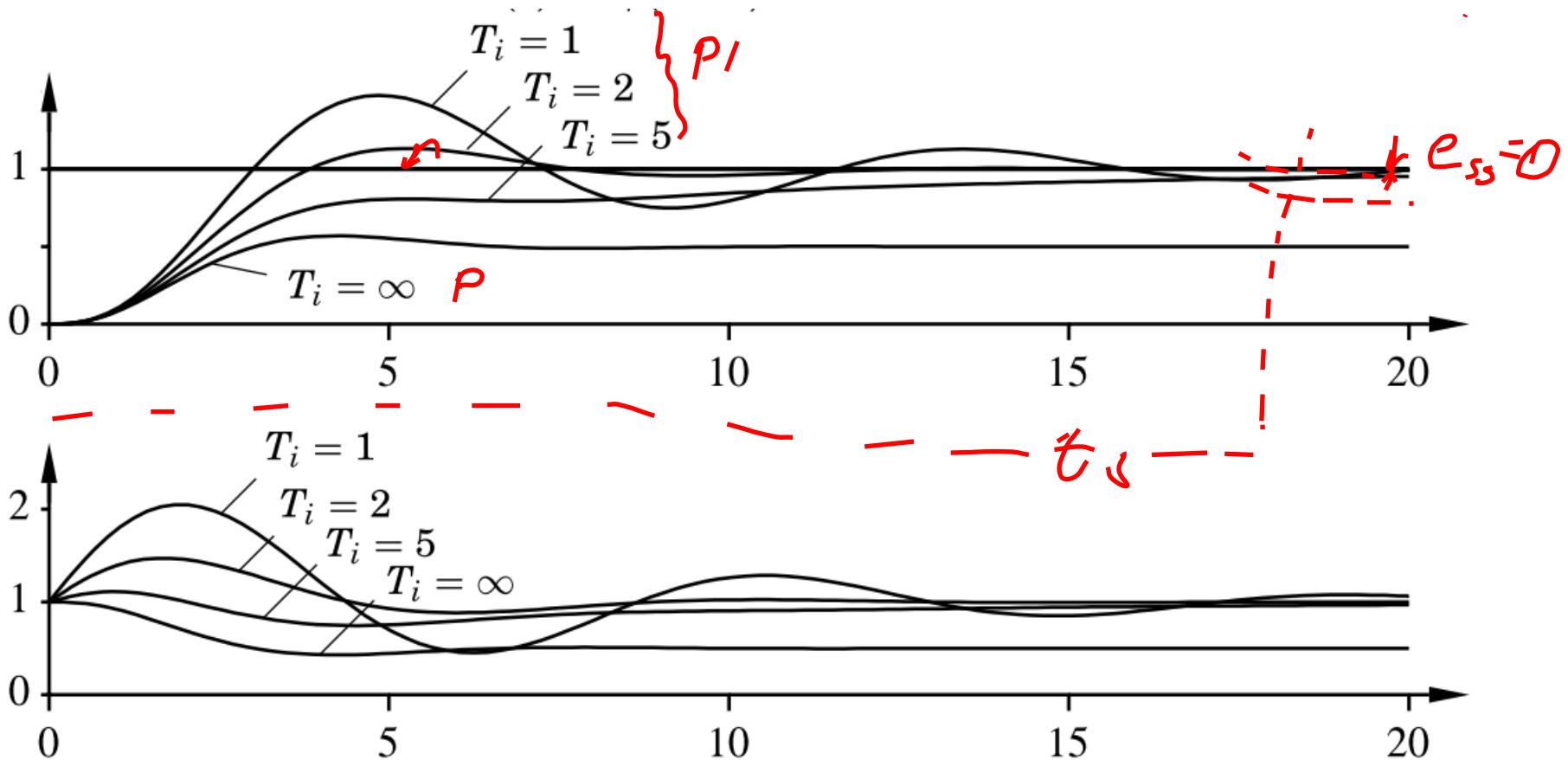


Reducing the rise time and steady-state error
 Increasing the oscillatory character

PI Controller

$K=1$

$$G_c = k \left(1 + \frac{1}{T_i s} \right)$$

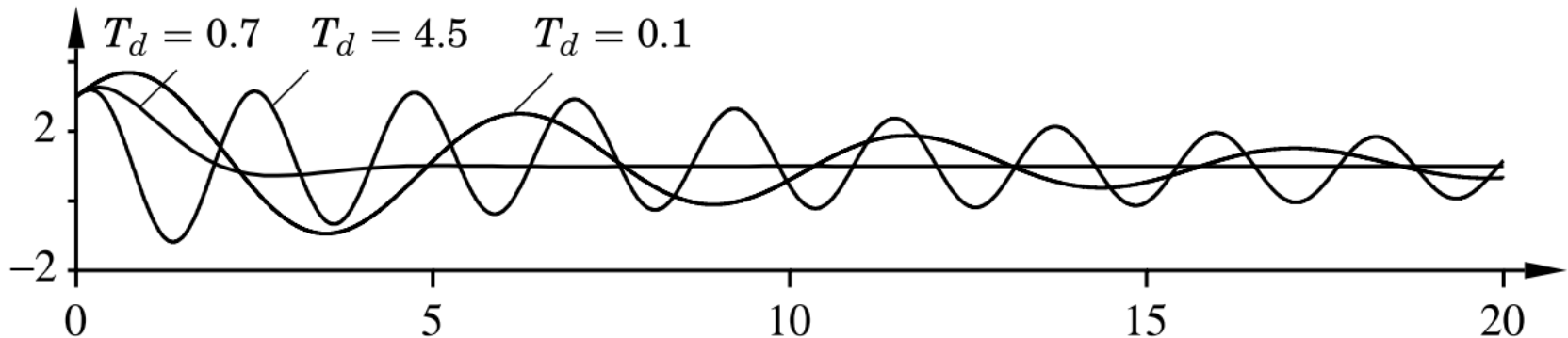
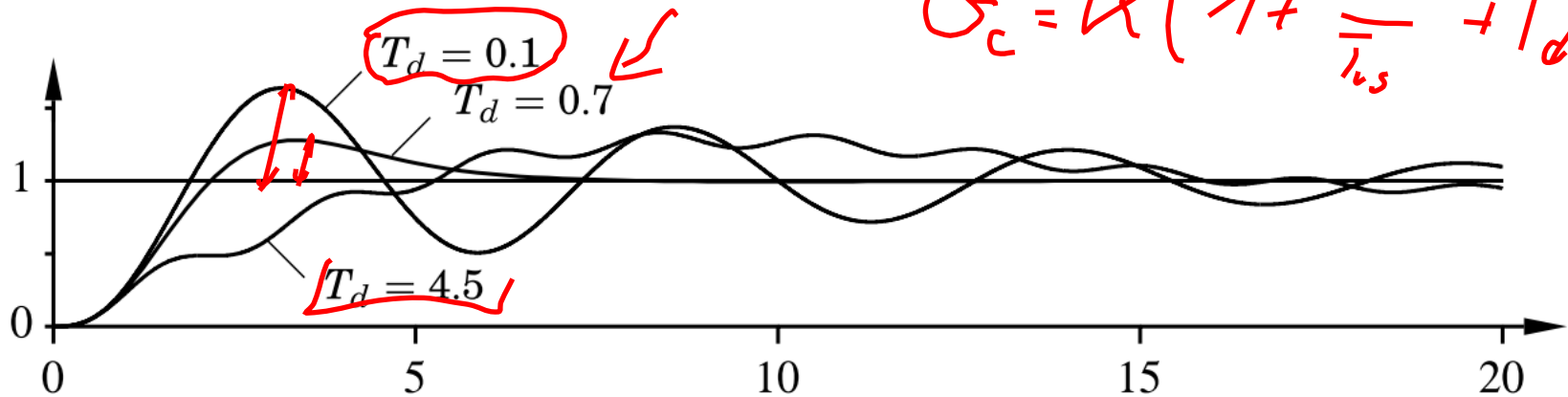


Elimination of the steady-state error
Increasing the oscillatory character

PID Controller.

K=3, Ti=2

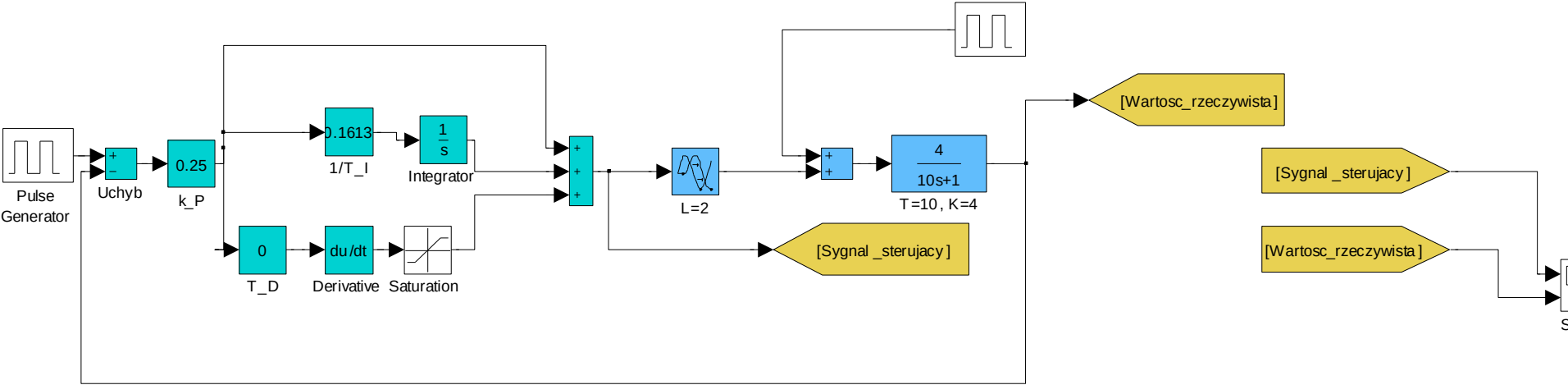
$$G_c = K \left(1 + \frac{1}{T_i s} + T_d s \right)$$

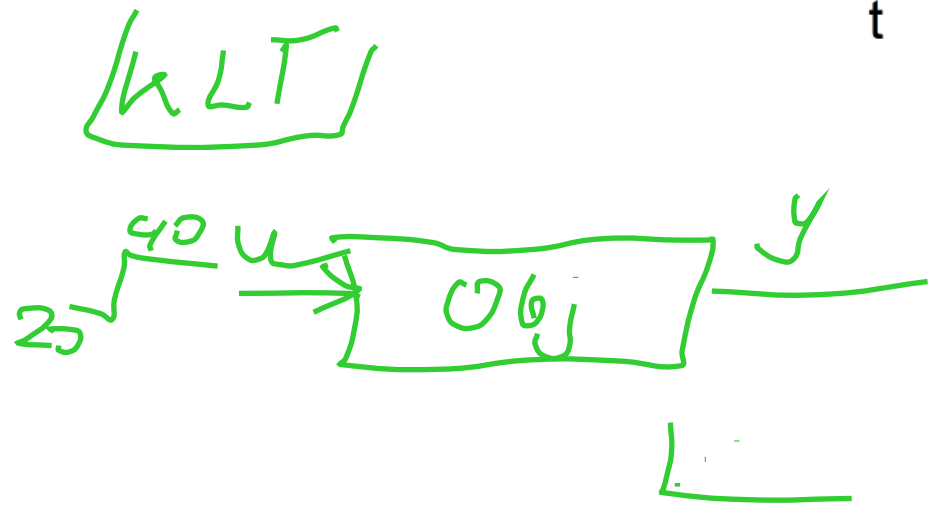
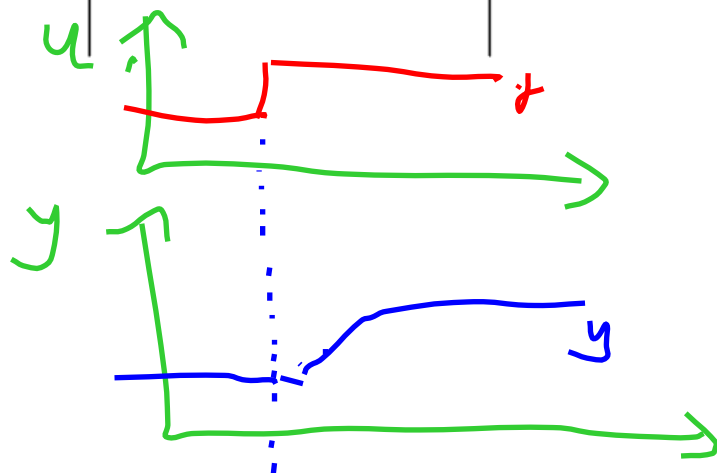
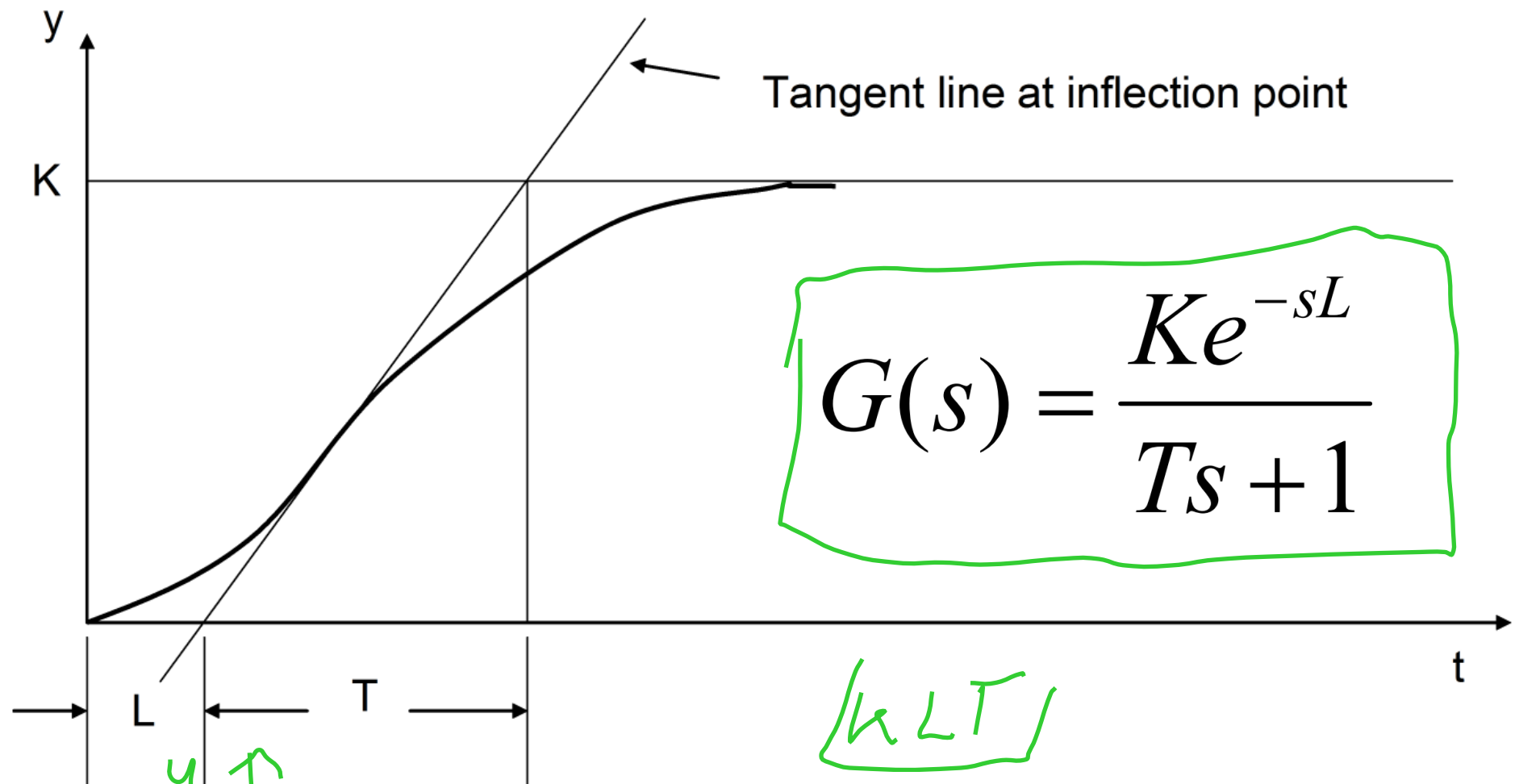


Decreasing but for larger T_d increasing
the oscillatory character

Response	Rise Time	Overshoot	Settling Time	S-S Error
K_P	Decrease	Increase	NT	Decrease
$K_I = \frac{k}{T_i}$	Decrease	Increase	Increase	Eliminate
$K_D = T_d \cdot k$	NT	Decrease	Decrease	NT

$K_P = k$
 $K_I = \frac{k}{T_i}$
 $K_D = k \cdot T_d$



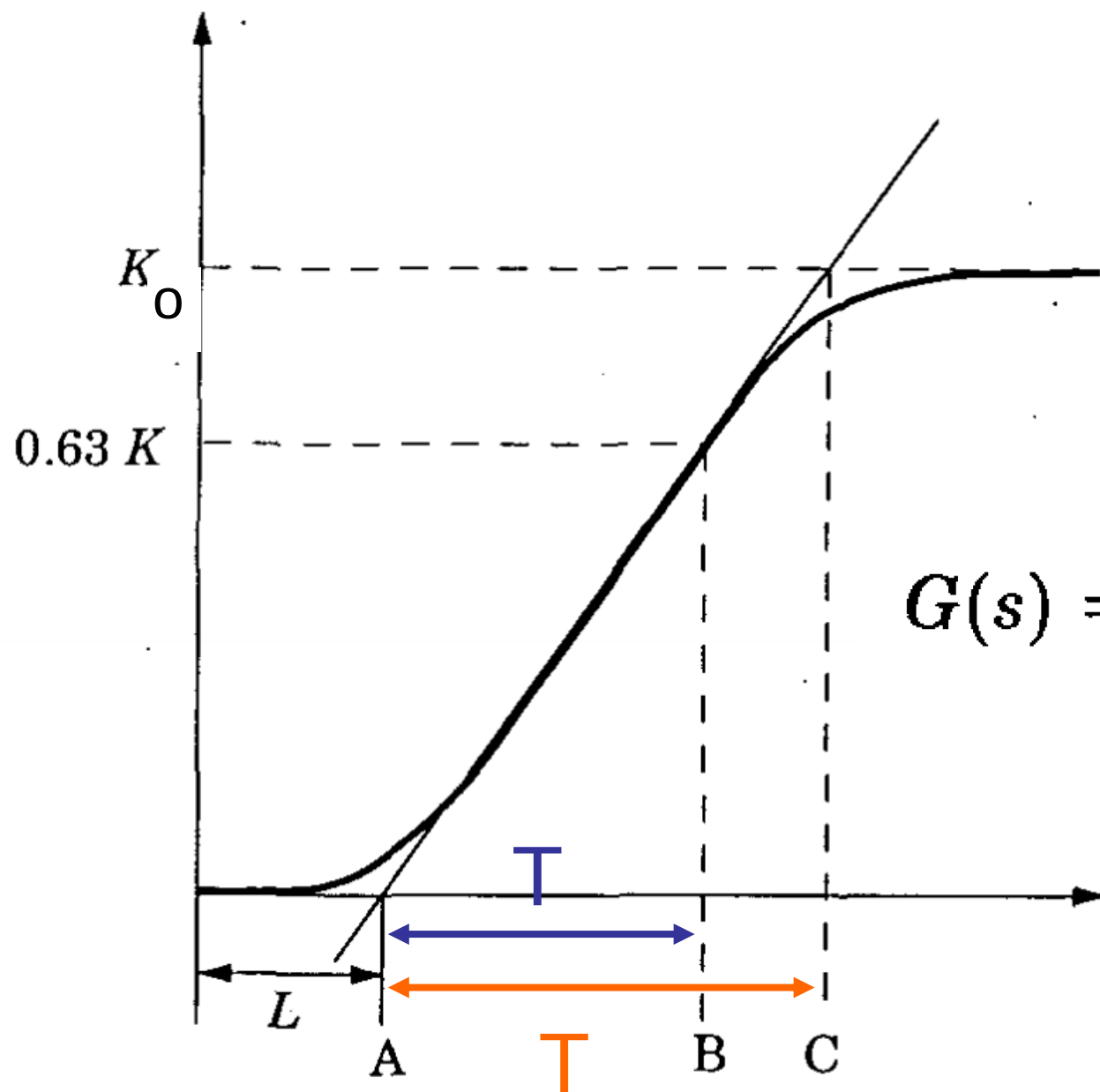


Modified step response method

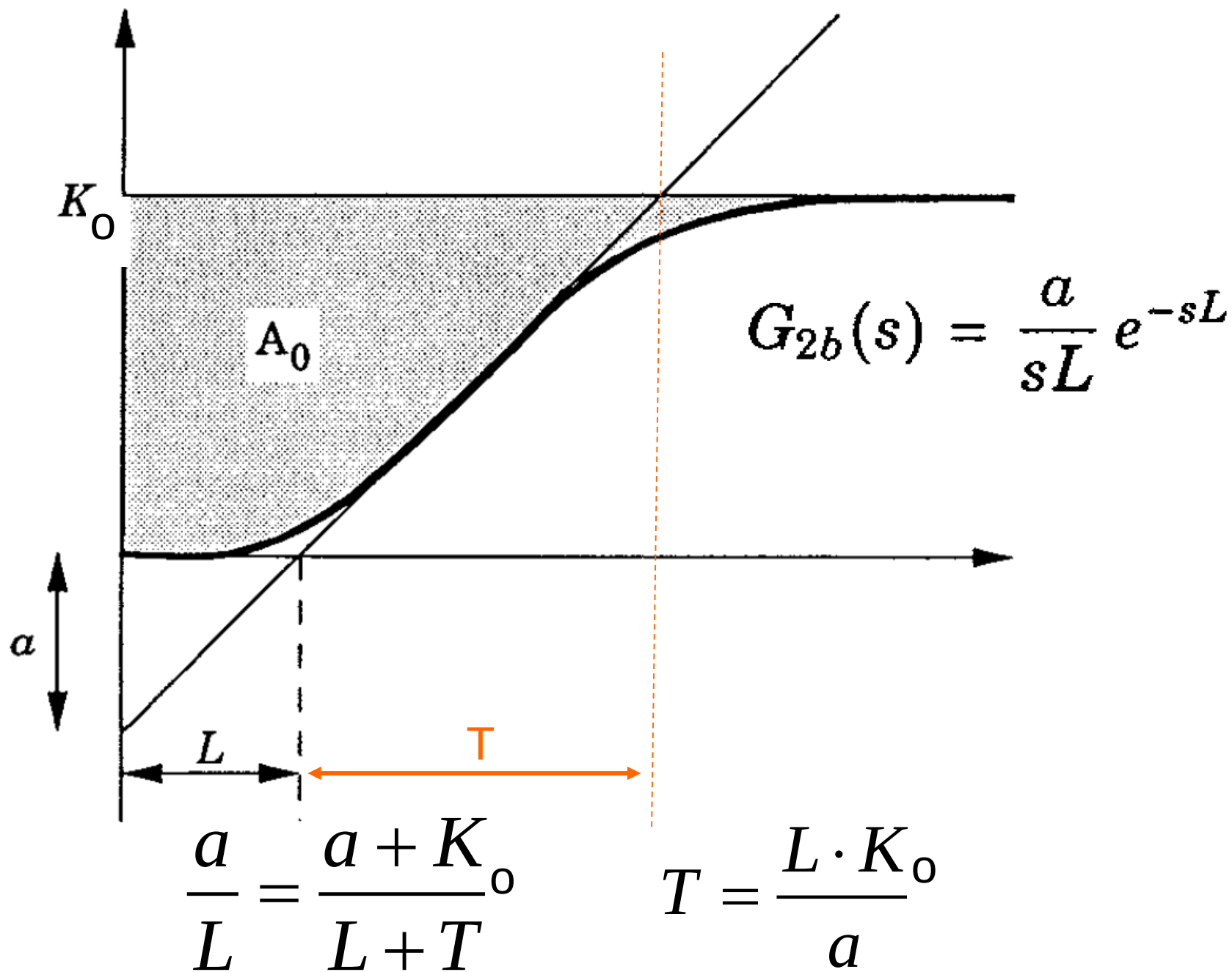
$$P_1(s) = \frac{k_p}{1 + sT} e^{-sL}$$

$$P_2(s) = \frac{k_v}{s} e^{-sL}$$

$$\tau = \frac{L}{L + T} \quad \text{relative time delay.}$$



$$G(s) = \frac{K_0}{1 + sT} e^{-sL}$$



The Ziegler-Nichols Tuning Rule Table

Controller	K	T_i	T_d	T_p
P	$1/a$			$4L$
PI	$0.9/a$	$3L$		$5.7L$
PID	$1.2/a$	$2L$	$L/2$	$3.4L$

$$u(t) = K \left(e(t) + \frac{1}{T_i} \int_0^t e(\tau) d\tau + T_d \frac{de(t)}{dt} \right)$$

$$u(t) = K \left(e(t) + \frac{1}{T_i} \int_0^t e(\tau) d\tau + T_d \frac{de(t)}{dt} \right)$$

PID Type	K^*K_p	$T_i=K_p/K_i$	$T_d=K_d/K_p$
P	$\frac{T}{L}$	∞	0
PI	$0.9 \frac{T}{L}$	$3L$	0
PID	$1.2 \frac{T}{L}$	$2L$	$0.5L$

Frequency method ZN

- 1) Select the controller type P
- 2) Slowly increase the gain until the output signal continues to oscillate.
- 3) Keep the value of the gain - K_u and the period of oscillation T_u

Selection of settings for the frequency method

Controller	K	T_i	T_d	T_p
P	$0.5K_u$			T_u
PI	$0.4K_u$	$0.8T_u$		$1.4T_u$
PID	$0.6K_u$	$0.5T_u$	$0.125T_u$	$0.85T_u$

Comparison of step-response and frequency response methods

$$G(s) = \frac{b}{s} e^{-sL}$$

	frequency	Step response
PI	$K = \frac{0.63}{bL}, \quad T_i = 3.2L$	$K = \frac{0.9}{bL}, \quad T_i = 3L$
PID	$K = \frac{0.94}{bL},$ $T_i = 2L \quad \text{and} \quad T_d = \frac{L}{2}$	$K = \frac{1.2}{bL},$

Critical evaluation of the ZN method

Relative deadlock time: $\tau = L/(L+T)$

- $\tau < 0.15$ - easy case, dominated by inertia -

As a rule, a PI controller is sufficient. Part D can improve dynamics, especially when the impact of disturbances is small. The ZN-setting can be clearly improved by other criteria. The ZN method does not describe the selection of the controller of the PD type, which may be beneficial in this case.

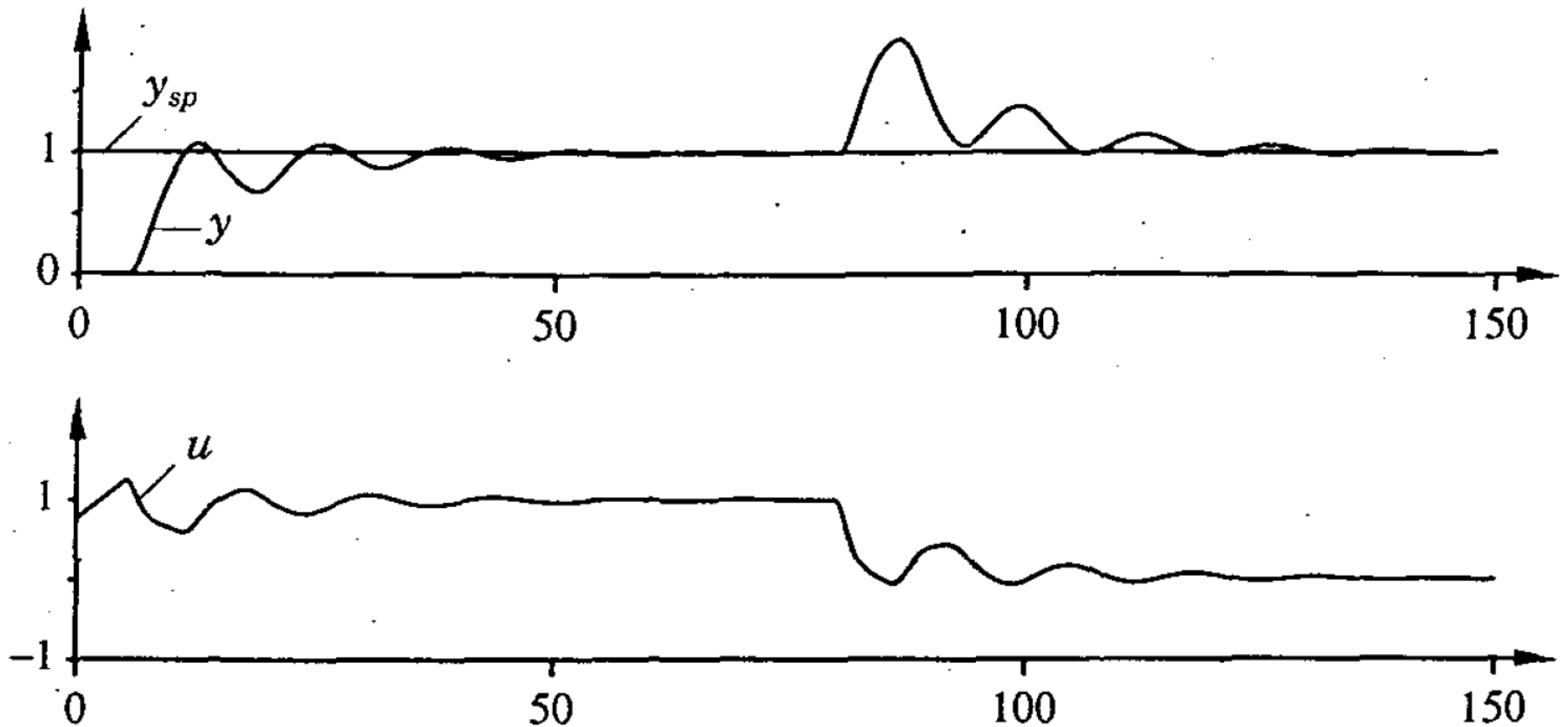
Critical evaluation of the ZN method

- $0.15 < \tau < 0.40$ - medium case. The ZN method gives good results in terms of response to disturbance, but large over-regulation in response to a change in the setpoint signal (attenuation $1/4$). It is advisable to use weighting factors (b, c) in the setpoint signal path.

Critical evaluation of the ZN method

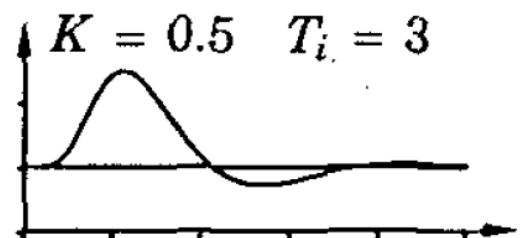
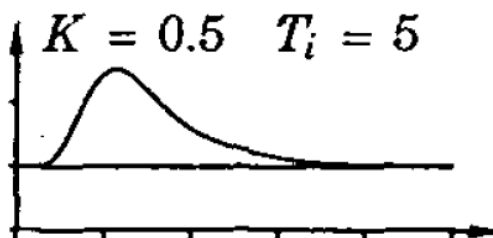
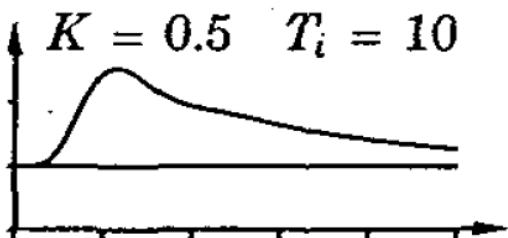
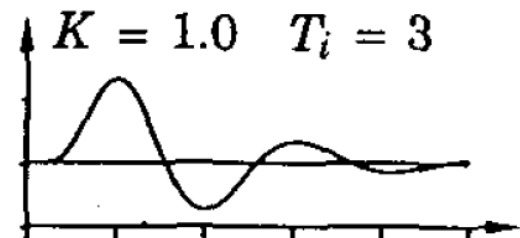
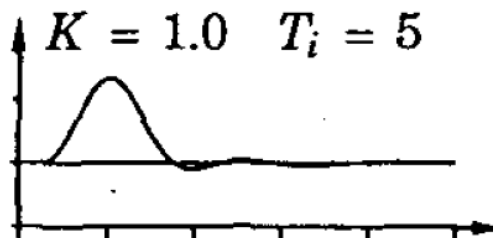
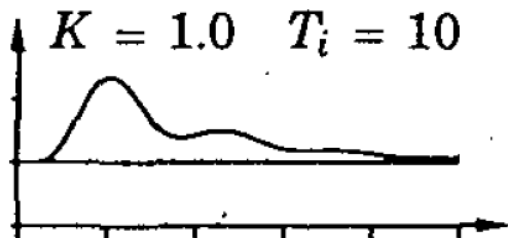
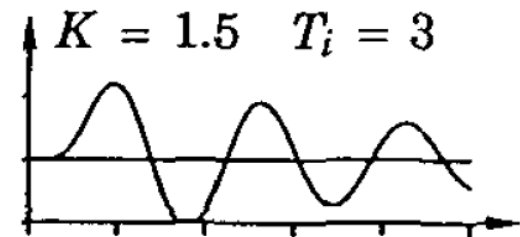
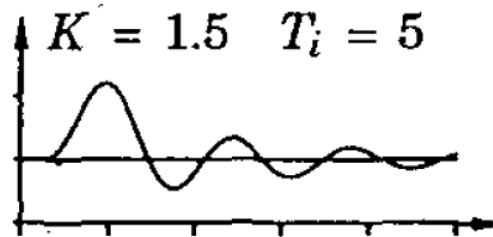
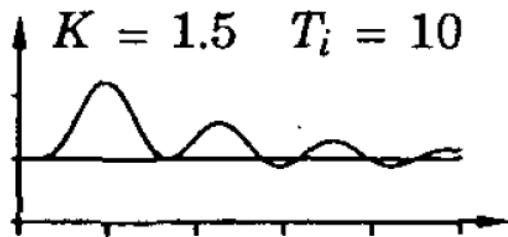
- $\tau > 0.40$ - case difficult, with dominant delay. The ZN method gives poor results. The PI/PID controller can be used, but requires different selection criteria. Advanced structures, such as the Smith predictive method, can provide a significant improvement.

PID controller for object with dominant delay $\tau=0.65$



$$G(s) = e^{-5s} / (s + 1)^3$$

Experimental tuning of the controller



The Chien, Hrones and Reswick Method (CRH)

- Provides better damping than ZN
- Signal waveforms without over-regulation or with 20% overregulation
- Other criteria for step response optimisation, other criteria for disturbance response

CHR method - response to disturbance

Overshoot		0%			20%		
Controller	K	T_i	T_d		K	T_i	T_d
P	$0.3/a$				$0.7/a$		
PI	$0.6/a$	$4L$			$0.7/a$	$2.3L$	
PID	$0.95/a$	$2.4L$	$0.42L$		$1.2/a$	$2L$	$0.42L$

$$T = \frac{L \cdot K_o}{a}$$

CHR method - response to setpoint step change

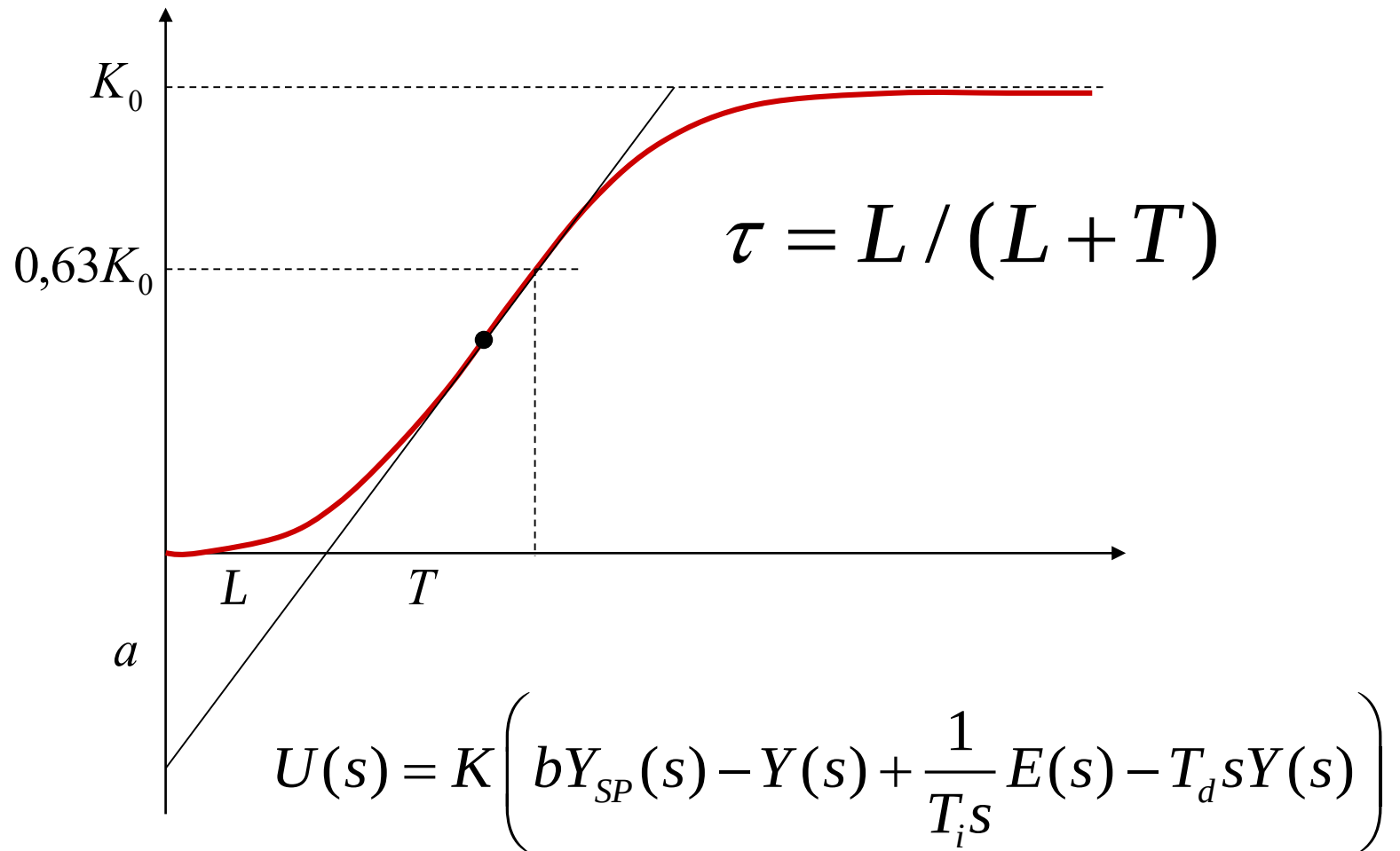
Overshoot	0%			20%		
Controller	K	T_i	T_d	K	T_i	T_d
P	$0.3/a$			$0.7/a$		
PI	$0.35/a$	$1.2T$		$0.6/a$	T	
PID	$0.6/a$	T	$0.5L$	$0.95/a$	$1.4T$	$0.47L$

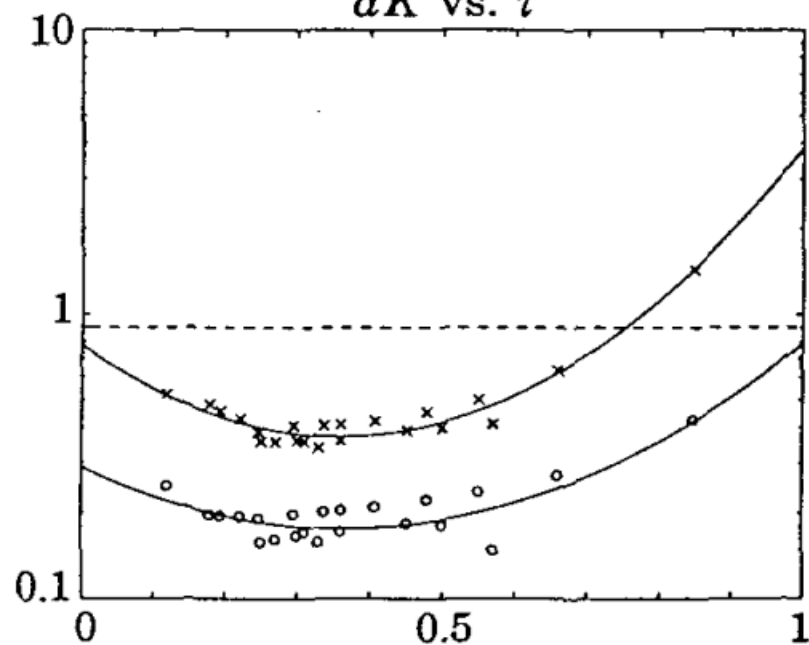
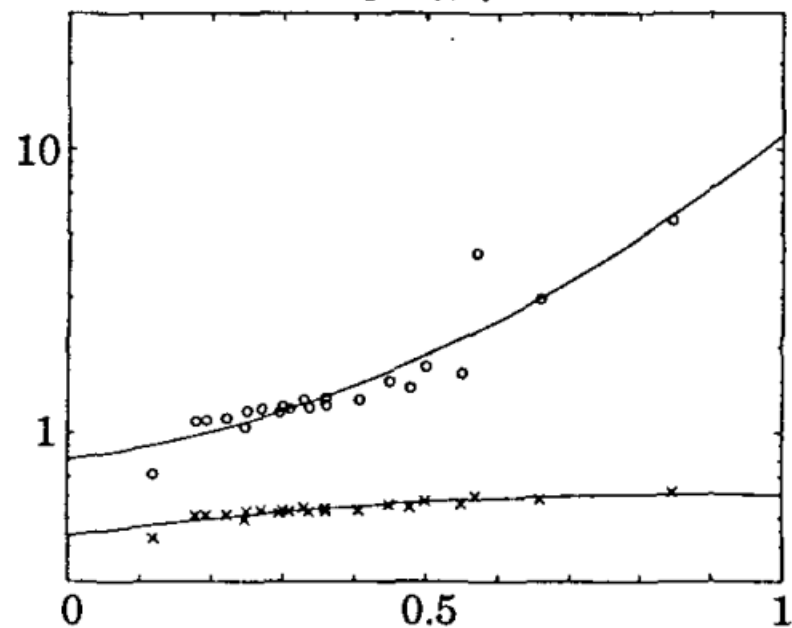
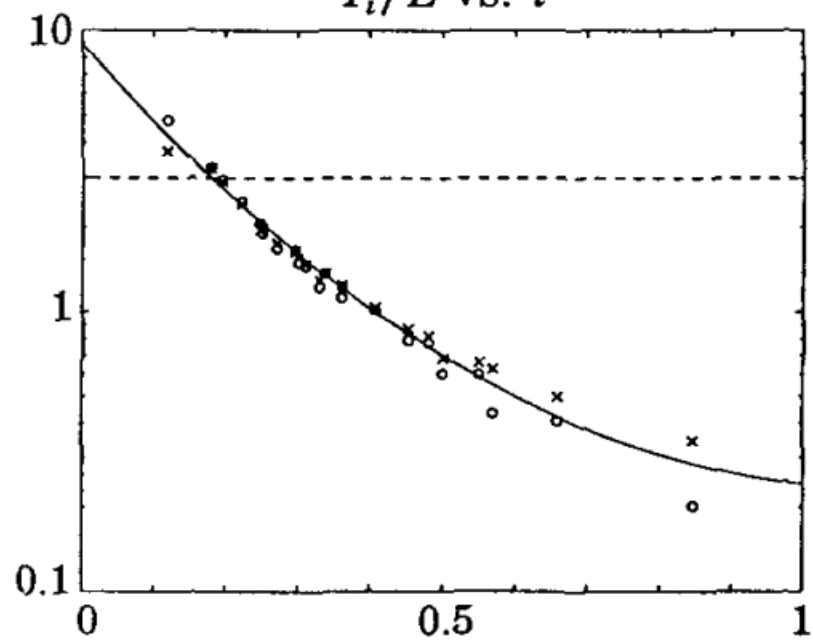
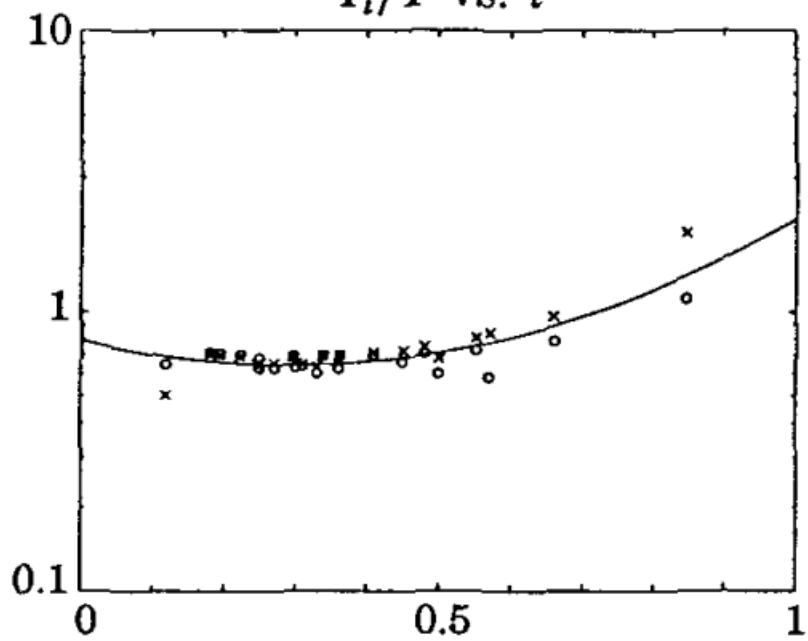
$$T = \frac{L \cdot K_0}{a}$$

Astrom's method

- Developed on the basis of experimental studies of a wide range of objects
- Calculations - based on the jump response
- as in the ZN method
- It takes into account the structure of the controller with a weighting factor of b
- For sensitivity parameter $M=1,4$ or $M=2$

Astrom, Hagglund (1995)



aK vs. τ  b vs. τ  T_i/L vs. τ  T_i/T vs. τ 

Astrom method - PI controller

$$f(\tau) = a_0 \exp(a_1 \tau + a_2 \tau^2)$$

	$M_s = 1.4$			$M_s = 2.0$		
	a_0	a_1	a_2	a_0	a_1	a_2
aK	0.29	-2.7	3.7	0.78	-4.1	5.7
T_i/L	8.9	-6.6	3.0	8.9	-6.6	3.0
T_i/T	0.79	-1.4	2.4	0.79	-1.4	2.4
b	0.81	0.73	1.9	0.44	0.78	-0.45

Astrom method - PID controller

$$f(\tau) = a_0 \exp(a_1 \tau + a_2 \tau^2)$$

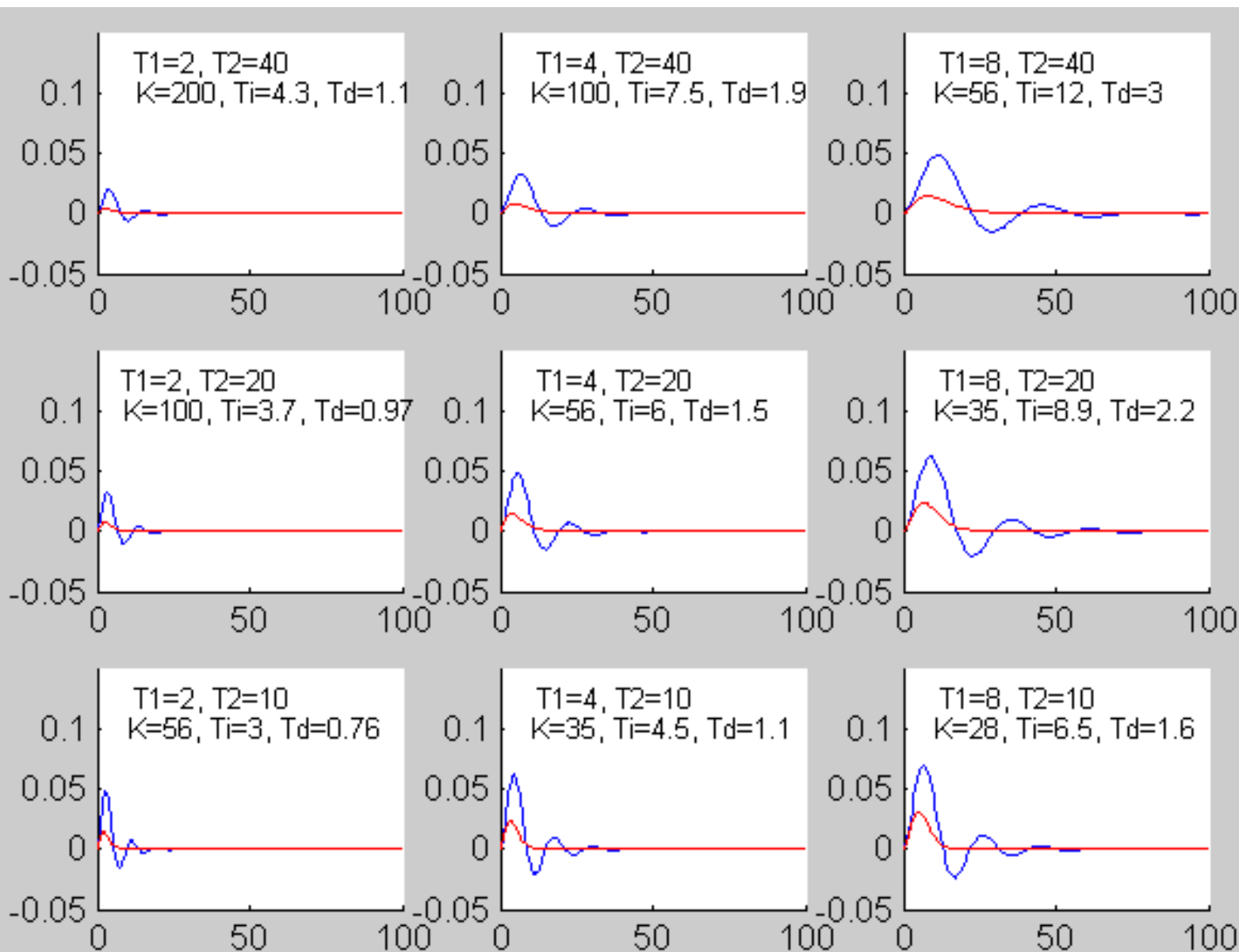
$$M_s = 1.4$$

$$M_s = 2.0$$

	a_0	a_1	a_2	a_0	a_1	a_2
aK	3.8	-8.4	7.3	8.4	-9.6	9.8
T_i/L	5.2	-2.5	-1.4	3.2	-1.5	-0.93
T_i/T	0.46	2.8	-2.1	0.28	3.8	-1.6
T_d/L	0.89	-0.37	-4.1	0.86	-1.9	-0.44
T_d/T	0.077	5.0	-4.8	0.076	3.4	-1.1
b	0.40	0.18	2.8	0.22	0.65	0.051

$$F(s) = \frac{1}{(T_1s + 1)(T_2s + 1)}$$

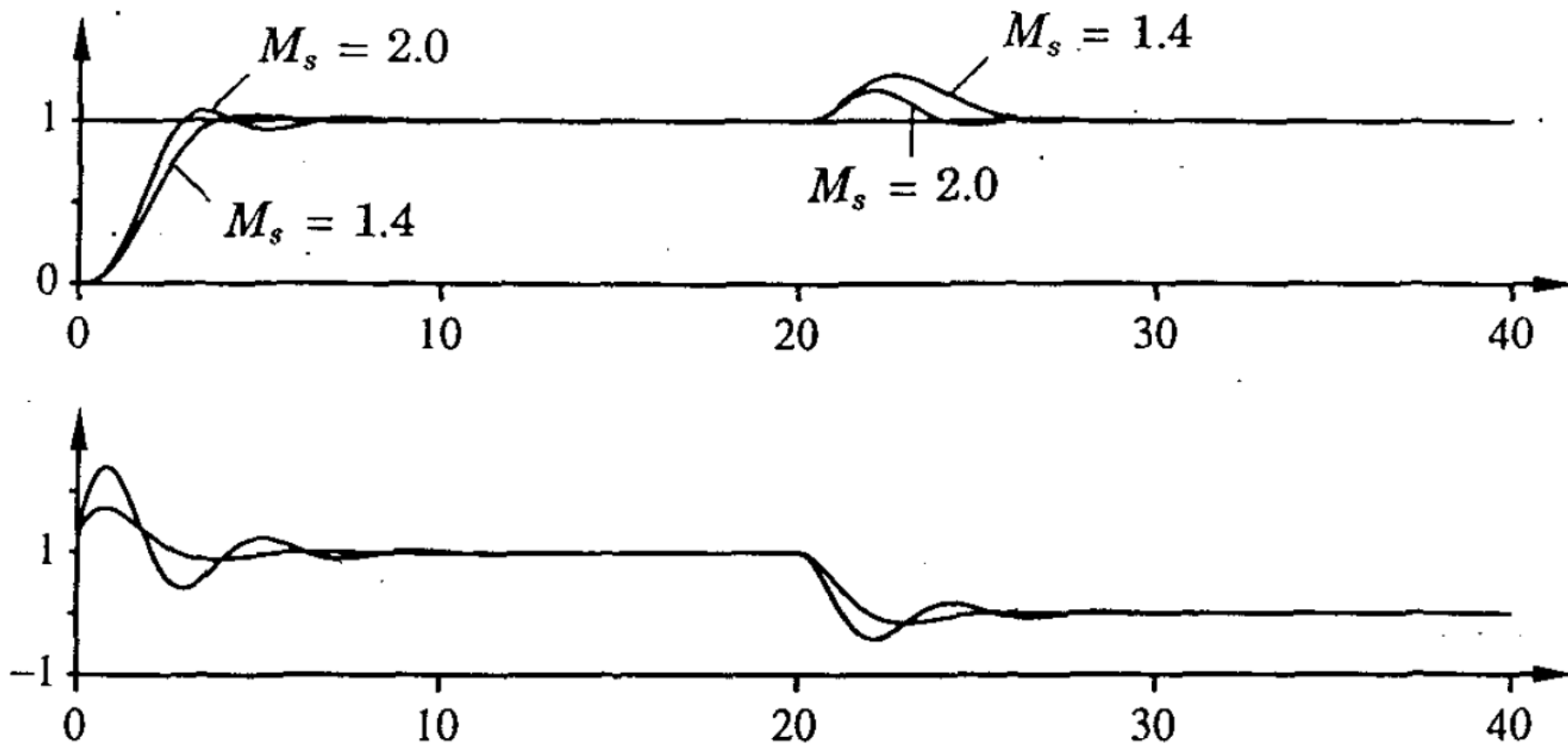
— Astrom-Hagglund
— Ziegler-Nichols



	$M_s = 1.4$	$M_s = 2.0$
K	2.5	4.8
T_i	2.2	1.8
T_d	0.56	0.46
b	0.52	0.27

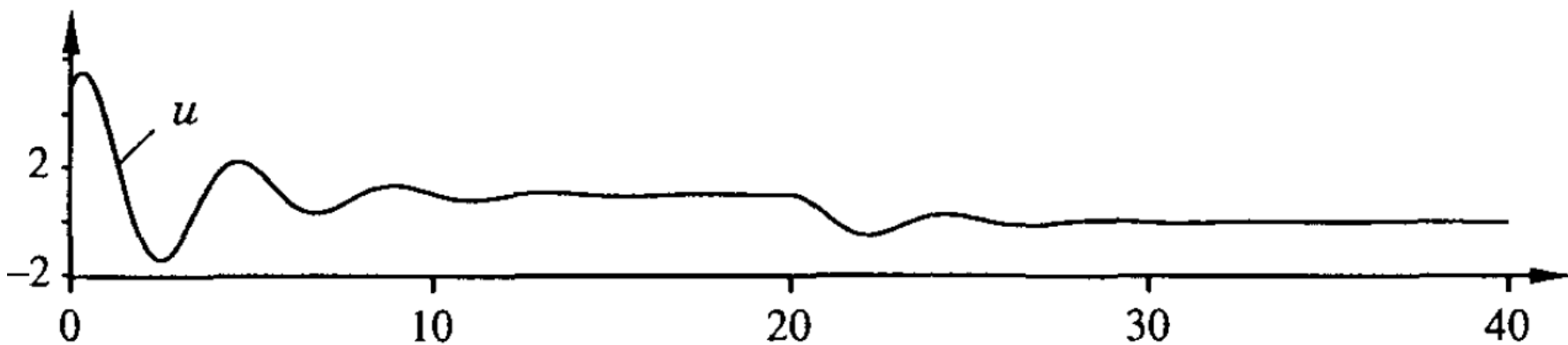
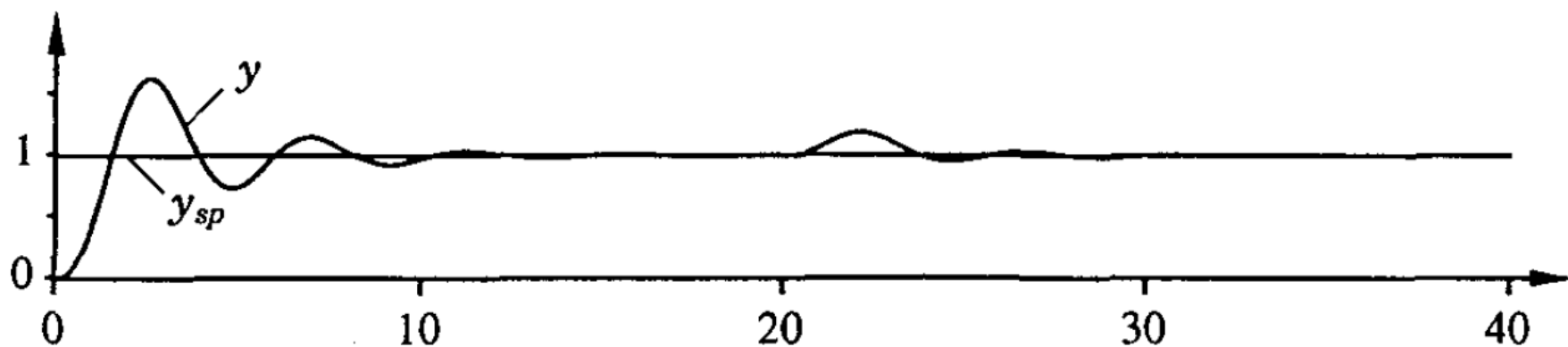
Astrom's method

$$1/(s+1)^3$$



ZN method

$$1/(s+1)^3$$



Simplified Astrom method $G_o(s) = \frac{K_o}{1 + sT} e^{-sL}$

$$K = \frac{1}{K_o} \left(0.2 + 0.45 \frac{T}{L} \right)$$

$$T_i = \frac{0.4L + 0.8T}{L + 0.1T} L$$

$$T_d = \frac{0.5LT}{0.3L + T}$$

$$\tau = L / (L + T)$$

$$b = \begin{cases} 0 & \text{for } \tau \leq 0.5 \\ 1 & \text{for } \tau > 0.5 \end{cases}$$

For an object with a finite response to a control signal step change

Simplified Astrom
method

$$G_o(s) = \frac{K_v}{s} e^{-sL}$$

$$K = 0.45/K_v$$

$$T_i = 8L$$

$$T_d = 0.5L$$

$$b = \begin{cases} 0 & \text{for } \tau \leq 0.5 \\ 1 & \text{for } \tau > 0.5 \end{cases}$$

For an integrating object